

(Un)Natural Language: An Art Software Identifies Potential Ecological Threats in Documents

Peijing Mou
Design Media Arts
University of California Los Angeles
Los Angeles, CA, USA
art@melodymou.com

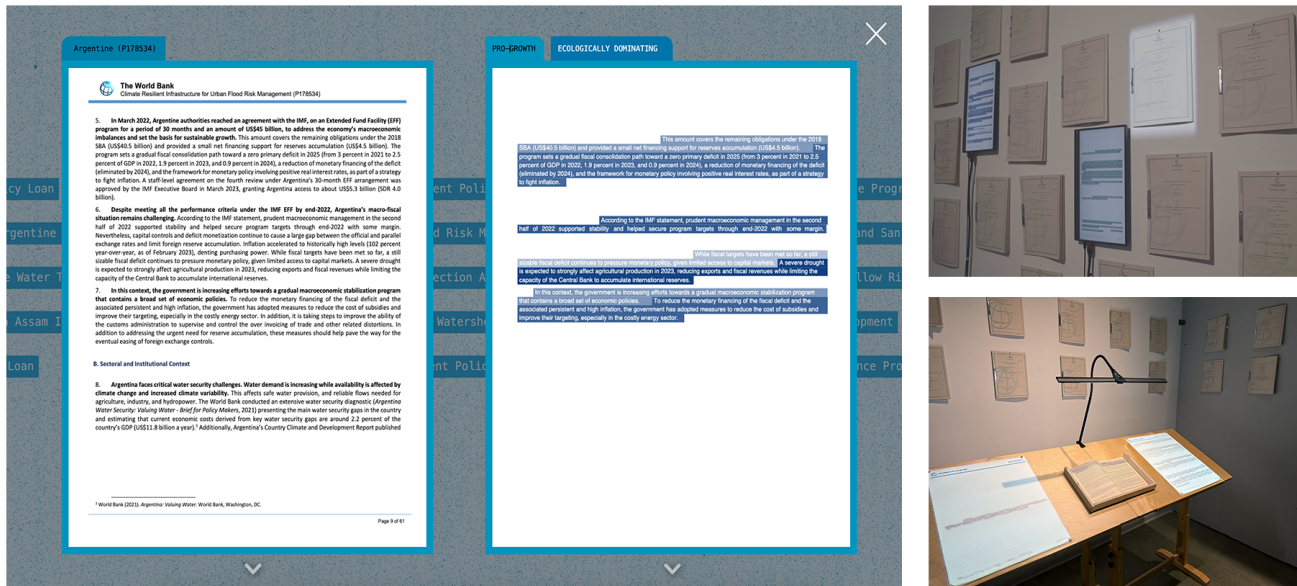


Figure 1: Online archive analysis page and two installations developed from the art software. (©Peijing Mou.)

Abstract

This paper introduces *(Un)Natural Language*, an artistic computational system that examines how words shape ecological narratives, detecting potential ecological threats in government water-related project documents. Weaving together machine learning, environmental activism, and linguistics, the system offers a new analytical lens for interpreting public documents, uncovering hidden narratives of economic expansion and extractivism. Through a custom-labeled dataset and a fine-tuned BERT model, it reveals and visualizes patterns of pro-growth and ecologically detrimental discourse. Developed into an interactive online archive and installation series, the project reclaims computation as a space for reflection rather than control—inviting viewers to rethink the language shaping our ecological futures.



This work is licensed under a Creative Commons Attribution 4.0 International License.
SA Art Papers '25, Hong Kong, Hong Kong
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2129-8/25/12
<https://doi.org/10.1145/3757369.3767611>

CCS Concepts

• Applied computing → Media arts; • Computing methodologies → Natural language processing;

Keywords

Machine Learning, Ecolinguistics, Degrowth, Net Art, Art-based Research

ACM Reference Format:

Peijing Mou. 2025. (Un)Natural Language: An Art Software Identifies Potential Ecological Threats in Documents. In *SIGGRAPH Asia 2025 Art Papers (SA Art Papers '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3757369.3767611>

1 Introduction

What narratives lie buried beneath the gridlines of urban development, within the margins of policy and loan documents, or in the silences surrounding ecological impact? What stories are concealed beneath the public language of government, in service of economic growth? *(Un)Natural Language* is an art software, further developed into an interactive online archive and installations, that examines and documents how words make worlds in government project

documents. It is an experimental project exploring the intersection of computational technology, ecology and linguistics to create a new analytical lens for complex issues in urbanization related to water, rivers and oceans.

This work offers an alternative application of computing, creating new ecological discourses that critically reflect on water urbanization. The collected government documents form a sequence-based dataset, which is labelled with three specific categories—growth, ecologically dominating and neutral. A language model is subsequently trained on this dataset to detect sentences that embody growth-oriented and extractive ideologies. In doing so, it reveals the hidden assumptions from the government texts, tracing the linguistic fingerprints of ecological narratives embedded in urban water governance, management and development.

As Donna Haraway writes, “it matters what stories we tell to tell other stories with; . . . It matters what stories make worlds; what worlds make stories.” Inspired by this, this project expands the possibilities for public engagement with language as an active ecological force, illustrating how words can shape, distort, and potentially heal our relationship with the environment [Haraway 2016]. This article presents a working ecological natural language processing (NLP) system that operates both as a critical research tool and as an artistic presentation for gallery audiences, contributing to ecology environmental activism, net art, and alternative NLP. By lifting the veil of bureaucratic language, it empowers viewers and users to recognize, question, and reimagine the hidden ideological structures shaping ecological discourse.

2 Background and Related Work

2.1 Ec linguistics and Natural Language Processing

Ecological linguistics, or ecolinguistics, expands sociolinguistics by situating language within broader ecological contexts—bridging social, environmental, and interspecies dimensions of discourse. A key practice within this field is eco-critical discourse analysis (eco-CDA), which critically examines how language perpetuates or challenges ecologically harmful ideologies [Zhang et al. 2023]. Halliday (1990) was among the first to urge applied linguists to confront urgent ecological issues, identifying “economic growth” narratives as central culprits: countless global texts, he noted, reinforce the notion that “growth is good, bigger is better,” normalizing ecological degradation [Halliday 1990]. Building on this, Stibbe (2015) developed an analytical framework intertwining language critique with ecological consciousness, emphasizing how linguistic patterns shape environmental imaginaries and contribute to either harm or healing [Stibbe 2015].

Currently, ecolinguistics research primarily focuses on small-scale, manual critical discourse analyses of texts, often requiring deep domain expertise and significant time investment. While essential for a nuanced understanding of texts, this manual method limits the scale and speed at which ecological narratives in large institutional documents can be systematically identified and challenged.

In parallel, computational methods such as NLP and Machine Learning (ML) have rapidly evolved, offering powerful tools for

language generation (e.g., ChatGPT), translation (e.g., Neural Machine Translation), classification, and summarization. However, in mainstream narratives, these technologies are overwhelmingly framed as instruments for development, optimization, and governance efficiency. Research directions predominantly focus on using ML to predict extreme weather events, optimize urban infrastructures, and manage natural resources. Phrases like “AI optimizes resource use,” and “ML enables smart governance” have become common, embedding a technocratic logic that frames natural and social systems as entities to be forecasted, managed, and optimized.

Despite their technical promise, these computational approaches often reinforce the ideologies that ecolinguistics seeks to critique. There is a notable gap in leveraging NLP and ML for critical ecological reading—to question the language of governance rather than serve it. Few projects attempt to operationalize eco-critical discourse analysis at scale, combining close reading sensitivity with the speed and scale made possible by computational tools.

2.2 Related Work

To resist this tendency, artists working at the intersection of technology and ecology have developed critical practices that subvert the dominant narratives of data and computation. Since the 1990s, artists have been reimagining technical systems as platforms for ecological and political intervention. Natalie Jeremijenko’s participatory environmental work pioneered the use of digital tools and networked interactions to engage the public with issues of urban ecology and multispecies relations. More recent projects, such as Tega Brain’s Intergovernmental Panel on Capitalism (2015) and Synthetic Messenger (2021), continue this trajectory by intervening in digital infrastructures to reveal the hidden ecological and economic logics embedded in media systems [Brain 2021]. Anna Tsing’s Feral Atlas (2020) further expands this field by creating a collaborative, browser-based atlas that maps the entangled effects of capitalist infrastructure and ecological disturbance [Tsing et al. 2020]. This project extends these critical art practices by operationalizing ecological critique at scale, uniquely employing NLP for ecolinguistics analysis.

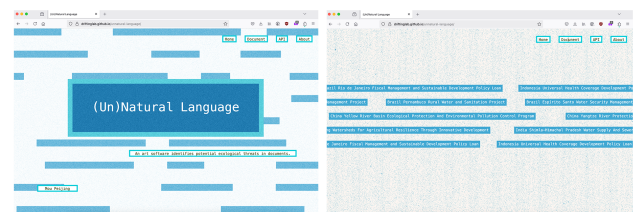


Figure 2: Online archive home page (left); archive page (right). (©Peijing Mou.)

3 (Un)Natural Language Project

3.1 Concept

(Un)Natural Language carries a dual meaning: it critiques the ideological constructions hidden within so-called “natural language,” while simultaneously referencing NLP technology. This project

collects program appraisal documents from the World Bank, which include program descriptions, implementation plans, frameworks, and assessments—particularly for hydrology-related projects such as water security and water management. A close reading of these documents reveals that the language employed is far from neutral; rather, it carries implicit ideological assumptions often overlooked in bureaucratic discourse. Specifically, it reflects a capitalist mindset that privileges economic expansion over relational forms of development or alternative ways of living. To excavate these hidden narratives, this project systematically labels each sentence into three categories: pro-growth, ecologically dominating, and neutral.

Growth, as Hickel [Hickel 2020] observes, is the ideological core of capitalism, driven by accumulation, commodification, and colonization. Growth hegemony refers to the cultural dominance of this ideology, often concealed within narratives of sustainability and technological progress. Pro-growth language demands increasing material and energy input, promotes economic expansion as inevitable, and frames technological innovation as the primary solution to ecological crises. Beyond surface promises, the construction of language reflects deeper anthropocentric and capitalist logics, subtly shaping public perception. Thus, critically identifying the ideological assumptions in public project documents is essential to resist the silent reproduction of growth-driven worldviews.

For instance, in the Viet Nam Binh Duong Province's Water Environment Improvement Project [for a Binh Duong Province's Water Environment Improvement Project 2023] (Figure 3), one sentence states: "This middle class has rising expectations... willingness to pay for improved municipal services..." The context reveals a market-based model of empowerment: participation and influence are linked to the ability to pay. This frames political agency as an extension of consumer power rather than a universal civic right, embedding capitalist governance logics within the language of environmental planning. Critically, from an eco-linguistics perspective, one must also interrogate the underlying capitalist values—such as equating middle-class expansion with societal advancement—that silently sustain and reproduce growth hegemony.

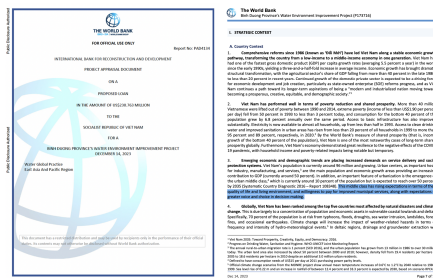


Figure 3: Viet Nam water environment improvement project document cover and page 7 from the World Bank. Highlight the analyzed sentence in blue. (©Peijing Mou.)

Ecological domination, on the other hand, reflects a worldview in which ecosystems, species, and landscapes are framed as passive, quantifiable resources to be managed, optimized, or extracted for human benefit [Plumwood 2002]. Rooted in critical theory, this

concept traces how Enlightenment rationality and capitalist development paradigms extend anthropocentric control into the ecological sphere. In contemporary discourse, ecologically dominating language often hides beneath narratives of development, management, and even conservation, framing human intervention as necessary and beneficial while perpetuating power over nonhuman life. Identifying ecological domination within language is crucial for unveiling how even well-intentioned environmental projects can replicate dominative attitudes toward nature, silencing its agency and complexity.

As seen in the same Viet Nam document: "With nearly 3,500 rivers ... the country is rich in water resources." This sentence exemplifies a common framing of nature as a stockpile of exploitable resources awaiting human utilization. Ecologically dominating language quantifies nature through metrics, distorting living ecosystems into extractable assets while erasing their agency and masking domination as management.

While pro-growth discourse emphasizes economic development, ecologically dominating language centers on extractivist logics. Though often intertwined, these two ideologies are distinct—some texts reflect both, while others reveal only one. Building on the critical definitions of pro-growth and ecologically dominating, this project repurposes an experimental computational model for ecological analysis. What *(Un)Natural Language* seeks to build a relational and ecological ML — powered by minimal, local energy, built on a system that supports a more-than-human world. It reconfigures algorithms to trace hidden ecological narratives embedded within urban development discourse.

3.2 Online Archive

Archive link: <https://driftinglab.github.io/unnatural-language/>

One artistic presentation developed from this software is an interactive web-based archive. Reflecting its thematic focus on water-related infrastructure, the interface adopts a soft, medium-light blue palette, evoking the fluidity of river flow.

Upon entering the platform, users can click either the title block or the rectangular "Document" tab to access the archive page (Figure 2), where blue blocks drift slowly across the screen, simulating the gentle movement of water. Each block represents a specific infrastructure project and contains both the original document and a visualized discourse analysis. Hovering pauses its movement; clicking opens the document analysis page (Figure 4), where the original World Bank document appears on the left, and the system-generated discourse visualization appears on the right. Users can further navigate between two analytical lenses — "Pro-Growth" and "Ecologically Dominating"—through tabs above. Highlighted sentences are aligned with their original positions, allowing for side-by-side comparison, and are color-coded according to the system's classification: darker hues signal explicit pro-growth or dominative logics, while lighter hues indicate more ambiguity.

Through this process, users are invited to reflect critically on how language subtly reinforces hegemonic relationships between human systems and the more-than-human world. The visualization makes the hidden ideological architectures visible. It serves as more than an illustrative tool; it functions as a critical intervention that

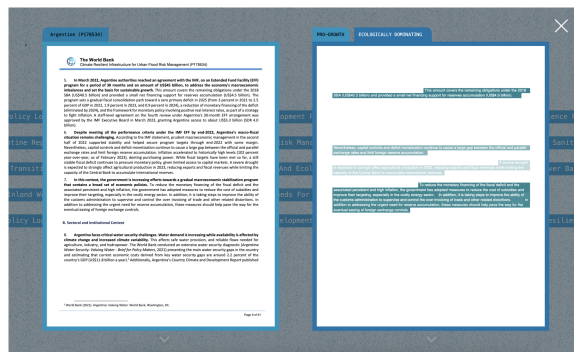


Figure 4: Online archive document analysis page, on the left is the original project document, on the right is the system-generated document. Here, the visualization on the right side shows the ecologically dominating language in the left document. (©Peijing Mou.)

reframes seemingly neutral bureaucratic language into contested terrains of meaning. By spatially mapping system-identified sentences alongside the original texts, the visualization disrupts the illusion of textual coherence and neutrality, rendering visible the fractures, emphases, and absences that structure environmental discourse. In this way, *(Un)Natural Language* not only reveals patterns otherwise imperceptible through traditional reading, but also foregrounds the politics of interpretation itself — inviting users to inhabit an active, interrogative stance toward documents that typically operate by naturalizing power.



Figure 5: API page (left) and About page (right). (©Peijing Mou.)

An essential component of this project is its API, which extends the platform’s functionality beyond the interface itself. *(Un)Natural Language* is conceived as a fully open-source, decentralized art software system, inviting artists, researchers, and technologists to engage with the work critically and collaboratively. In contrast to the closed, proprietary practices of industrial AI—where models and datasets are often hidden behind corporate walls—this project foregrounds technical legibility and ethical openness. The API page (Figure 5) allows users to input custom sentences and test the model online. All code and model infrastructure are available on GitHub, supporting both transparency and potential future development. The About page (Figure 5) provides additional context, including a critical glossary that outlines the concept and a brief introduction.

GitHub link: <https://github.com/DriftingLab/unnatural-language.git>

3.3 Installations

Two gallery installations were developed from the online archive for gallery audience, offering spatial interfaces for embodied engagement with the documents and machine detections. Each version creates a distinct viewing atmosphere, tailored to different audience scales and interaction rhythms.



Figure 6: Photographs of installation version 1. Audiences read and experience the work in the gallery. (©Peijing Mou.)

This version (Figure 6) adopts a more social format, inviting group interaction and casual exploration. It consists of a wall installation that arranges printed project documents and three displays in the form of a flowing river. All documents are hung on nails and can be physically removed, browsed, and compared by visitors. Two of the screens continuously display system-generated visualizations, organized under the pro-growth and ecologically dominating categories, cycling slowly page by page. The third screen presents interaction instructions and contextual background. A moving spotlight tracks the visualization timeline and highlights the corresponding printed document on the wall in real-time, prompting viewers to directly compare the physical and interpreted versions. The entire system—three monitors and a spotlight—is synchronized using MadMapper.



Figure 7: Photographs of installation version 2. The left one shows the document room. The right one focuses on the interaction table. (©Peijing Mou.)

Designed as an immersive and contemplative environment, the other version (Figure 7) uses a three-wall layout hung with printed documents, and a large, tilted table placed centrally, which is more suited to individual or small-group engagement. An idle projection on the table displays instructions and invites participation. Viewers

can pick a document from the wall and place it onto a central RFID-embedded platform on the table. This action triggers two synchronized projections on the left and right side of the table: the left visualizes sentences categorized as pro-growth, while the right reveals ecologically dominating discourse from the same document. Each document is tagged with an RFID label, enabling automatic recognition and cueing via MadMapper for responsive projection mapping.

4 Technical Description

4.1 System Overview

This system (Figure 8) is built on a pretrained BERT (Bidirectional encoder representations from transformers) model, fine-tuned specifically for a multi-label classification task on a custom-labeled dataset. It classifies sentences such as pro-growth, ecologically dominating and neutral. The model focuses on identifying ecology-, management-, economics-related phrases and hidden ideologies based on actions (e.g., development, extraction, expansion), perspectives (e.g., capitalism, domination, greenwash), and priorities (e.g., economic growth). The system includes four primary phases: data collection and processing, labeling, training and interpretive output generation.

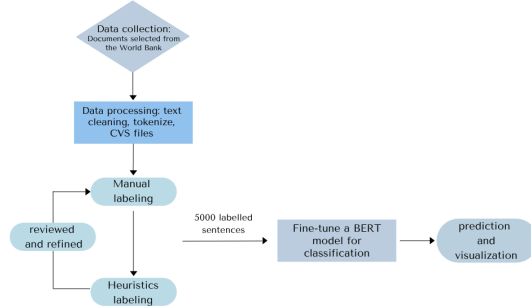


Figure 8: System workflow. (©Peijing Mou.)

4.2 Data and Labeling

Public documents from the World Bank were collected, cleaned, and segmented into sentences using the NLTK tokenizer in Python. Labeling was conducted on the Refinery platform using a hybrid, iterative human-machine workflow. Approximately 50% of the dataset was manually labelled to establish consistent criteria grounded in degrowth values. Sentences were marked as:

- **Pro-growth:** language that reflects a growth-oriented mindset, prioritizes economic expansion, or treats ecological and social limits as solvable via technological or market-based interventions.
- **Ecologically dominating:** language that frames nature through anthropocentric or managerial logics—treating ecosystems as resources to be controlled, extracted, or optimized.

These annotations informed rule-based heuristics that assisted in labeling the remaining data. The process was recursive: machine-generated labels were reviewed, refined, and selectively overwritten. In total, approximately 5000 sentences were labeled. The artist re-entered the loop multiple times to incorporate edge cases and update definitions—treating labeling as a critical intervention, not a neutral task.

4.3 Training

The labeled dataset was then exported to fine-tune a customized BERT model, modified with additional dropout and dense layers for multi-label text classification implemented in PyTorch. Once trained over 300 epochs, the model assigns category labels to new input sentences and returns a score for each labeled sentence based on its computed latent embeddings, indicating the degree to which the sentence aligns with the assigned category.

4.4 Output and Visualization

Taking the Morocco Water Security and Resilience Program (Report No: PAD5329) [for a water security and resilience program 2023] as a case study, the system first segments the 179-page document into individual sentences, each one based on its alignment with pro-growth or ecologically dominating discourse (Figure 9). Each sentence is assigned a label and a confidence score (0–1) indicating alignment strength.

Line	Original Text	Label	Confidence Score
1	After two decades of uninterrupted expansion the Moroccan economy is facing several challenges.	pro-growth	0.4118
2	Moroccan structural reforms launched in the late 1990s led to increase the country's real gross domestic product (GDP) by nearly 100 percent between 1999 and 2013, supported by capital flows from the Moroccan and IMF and a 10 percent increase in GDP, according to World Bank data. In the 2000s, the higher unemployment rate...	pro-growth	0.9999
3	After the gradual lifting of subsidies, economic growth began to pick up in the 2000s, and Morocco's real GDP rose by 1.5 percent in 2003. GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
4	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
5	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
6	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
7	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
8	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
9	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
10	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
11	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
12	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
13	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
14	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
15	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
16	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
17	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
18	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
19	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
20	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
21	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
22	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
23	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
24	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
25	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
26	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
27	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
28	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
29	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
30	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
31	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
32	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
33	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
34	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
35	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
36	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
37	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
38	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
39	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
40	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
41	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
42	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
43	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
44	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
45	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
46	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
47	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
48	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
49	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
50	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
51	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
52	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
53	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
54	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
55	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
56	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
57	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
58	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
59	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
60	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
61	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
62	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
63	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
64	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
65	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
66	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
67	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
68	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
69	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
70	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
71	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
72	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
73	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
74	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
75	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
76	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
77	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
78	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
79	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
80	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
81	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
82	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
83	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
84	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
85	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
86	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
87	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
88	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
89	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
90	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
91	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
92	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
93	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
94	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
95	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
96	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
97	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
98	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
99	The economic growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999
100	Real GDP growth slowed to 0.2 percent in 2002, reflecting the country's economic growth, leading to the country's economic growth...	pro-growth	0.9999

Figure 9: Original Morocco program document (top left); Segmentation result (top right); Returned labeled result with confidence scores from the system (bottom). (©Peijing Mou.)

Using the pro-growth category to explain how this system works, Figure 9. shows every pro-growth sentence in this document returned by the system. The degree of pro-growth alignment for each sentence is represented by a value between 0 and 1 in the right column. The first captured sentence “After two decades ... facing several challenges.” is scored at 0.4118 (rounded) with a pro-growth label. The phrase “uninterrupted expansion” naturalizes economic growth as a positive and inevitable trajectory, without questioning its structural assumptions. The hidden narratives behind give it a mediate value. Line 3, “Economic structural ... and almost

eradicate extreme poverty.”, returns 0.9749, showing an obvious growth-orientated tone. Visualized in Figure 10.1.

For line 15, “The New Development Model, ... progress and prosperity for all.”, from a linguistic angle, it uses positive word choices to convince the public that development is a linear and desirable process, foreclosing critical reflection on the socio-ecological consequences of accelerated expansion. This sentence is deeply entangled with a growth mindset which can be hard to detect, scored at a relatively low value of 0.1147. Visualization is a much light blue in Figure 10.2.

This visualization system turns the returns from the model into new generated documents. It gives the audience clear highlights to read and reflect the language used in these sentences, with different color shades indicating different levels of visibility and severity of growth or ecological domination. The bottom image in Figure 10.3 displays what the entire language pattern looks like.

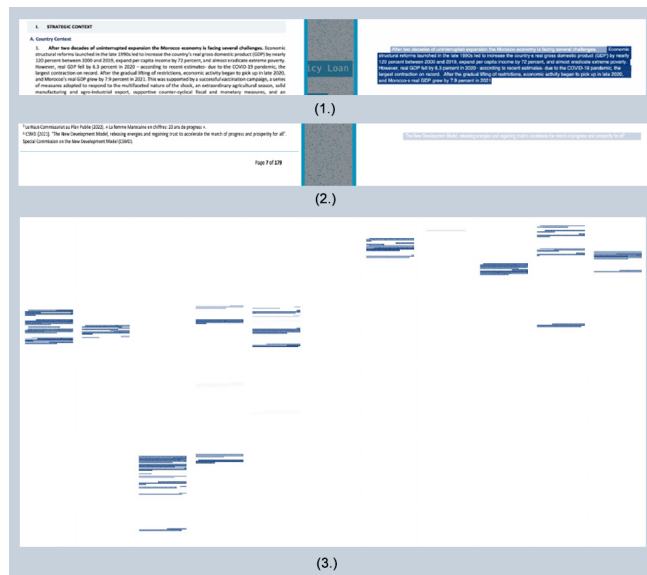


Figure 10: Model outputs visualization. Darker shades indicate stronger ideological alignment; New generated document overview. (©Peijing Mou.)

5 Contributions and Reflections

This project offers a new contribution at the intersection of climate activism, net art and alternative AI research. Within climate activism and public discourse, this project directly addresses the materiality of language—how words shape worlds. While conventional critical discourse analysis is often confined to academic contexts, this system not only increases the capacity for close and distinct reading without sacrificing critical sensitivity, but also significantly accelerates the process, making large-scale analysis feasible for broader publics.

Situated within net art, the model embraces the productive ambiguities of machine interpretation, rejecting the pursuit of technical perfection in favor of deeper political resonance. Unlike traditional computer science endeavors that pursue maximum model accuracy,

this system redefines effectiveness and success within an ecolinguistic frame: it treats returned values as the weight of ideological language rather than accurate classification prediction. The labeling process is inherently subjective—an intervention guided by the artist’s commitment to degrowth.

In the field of NLP, this project exemplifies an alternative use of AI: not to accelerate growth, optimize extraction, control or commodify attention, but to assist in building critical, slower, and more ecologically attuned reading practices. Technology is never neutral. Who designs it, for what purpose, and within which ideological framework determines whether it enables emancipation or domination. By situating itself in a degrowth-oriented framework and offering accessible tools for scholars, activists, local communities and the public, this project challenges the mainstream deployment of AI, suggesting new infrastructures for thought, resistance, and ecological care.

6 Conclusion

(Un)Natural Language is a critical artistic project that merges ecolinguistics and machine learning to analyze how institutional documents reproduce narratives of growth and ecological domination. Through a combination of data processing, automated sentence classification, and interactive visualization, it transforms bureaucratic language into ecological critique. It challenges the dominant technocratic narratives of pro-growth and offers an accessible, critical method for revealing ideologies within public discourse. As both a research tool and a public-facing artwork, it reclaims computation as a space for reflection rather than control—inviting users to rethink the language shaping our ecological futures.

Acknowledgments

The author gratefully acknowledges Chenyue xdd Dai, who contributes on the website development and visualization.

References

- Tega Brain. 2021. *Synthetic Messenger*. <https://syntheticmessenger.com>
- Viet Nam for a Binh Duong Province’s Water Environment Improvement Project. 2023. *Project appraisal document on a proposed loan to the Socialist Republic of Viet Nam for a Binh Duong Province’s Water Environment Improvement Project*. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099121923180526785>
- Morocco for a water security and resilience program. 2023. *Project appraisal document on a proposed loan to the Kingdom of Morocco for a water security and resilience program*. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099062823102563717>
- Michael Alexander Kirkwood Halliday. 1990. *New ways of meaning: The challenge to applied linguistics [Speech]*.
- Donna Jeanne Haraway. 2016. *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press.
- Jason Hickel. 2020. What does degrowth mean? A few points of clarification. *Globalizations* 18, 1 (2020), 1–7. doi:10.1080/14747731.2020.1812222
- Val Plumwood. 2002. *Environmental Culture: The Ecological Crisis of Reason*. Routledge.
- Arran Stibbe. 2015. *Ecolinguistics: Language, Ecology, and the Stories We Live By*. Routledge.
- Anna L. Tsing, Jennifer Deger, F. Keleman Saxena, and N. Zhou. 2020. *Feral Atlas: The More-than-Human Anthropocene*. <https://feralatlus.supdigital.org> Interactive website.
- Bing Zhang, Subramaniam Chitra Sandaran, and Jing Feng. 2023. The ecological discourse analysis of news discourse based on deep learning from the perspective of ecological philosophy. *PLoS ONE* 18, 1 (2023), e0280190. doi:10.1371/journal.pone.0280190